

Regular/Supplementary Summer Examination – 2025

Semester : V

Duration: 3 Hr.

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

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6	Which of the following is generally used to measure direct solar radiation? a) Pyranometer b) Actinometer c) Pyr heliometer d) IC tester	CO2	1
7	Which of the following is used to produce biogas from biomass? a) Anaerobic b) Aerobic c) Fermentation d) Pyrolysis treatment treatment	CO4	1
8	Which of the following is a mechanical application of wind energy? a) Lighting b) Electricity c) Pumping d) Wind water surfing	CO4	1
9	Which of the following is a part of a general wind energy conversion system? a) Server b) Wind turbine c) Cloud User d) Service Provider	CO4	1
10	How does a tidal lagoon work? a) By storing sea water b) By creating a difference in water levels c) By placing the turbine in the running water d) By creating barrages	CO5	1
11	_____ turbine is used in closed cycle ocean thermal energy conversion. a) Horizontal b) Low-pressure c) High-pressure d) Vertical	CO5	1
12	Which of the following is used as a moderator in a nuclear reactor? a) Cadmium b) Plutonium c) Uranium d) Heavy water	CO5	1
Q. 2 Solve the following.			12
A)	Differentiate between Renewable Energy sources and non-Renewable Energy sources.	CO1	6
B)	Explain in detail energy reserves in India.	CO1	6
Q.3 Solve the following.			12
A)	Define a) Declination angle b) Zenith angle c) Altitude angle.	CO2	6

B)	Explain with neat sketch construction and working of Flat plate collector. Also write its advantages and disadvantages.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	What is solar still? Explain solar distillation system with neat sketch.	CO3	6
B)	Explain any two types of solar concentrating collectors with neat sketch.	CO3	6
C)	What is PV cell? Explain how it converts solar energy into electrical energy.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	What is wind energy? Explain selection of site for wind turbine generation system.	CO4	6
B)	How the wind mills are classified? Sketch the diagram of a Horizontal Axis Wind mill, and explain the function of its main components.	CO4	6
C)	What is biomass energy? Explain with neat sketch floating drum type Biogas plant.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	What is OTEC system? Explain with neat sketch the working of open cycle OTEC system.	CO5	6
B)	Write a short note on 'Geothermal energy'.	CO5	6
C)	Draw neat sketch of Hydro-electric power plant. How the electric energy is generated in a hydroelectric power plant?	CO5	6

***** End *****